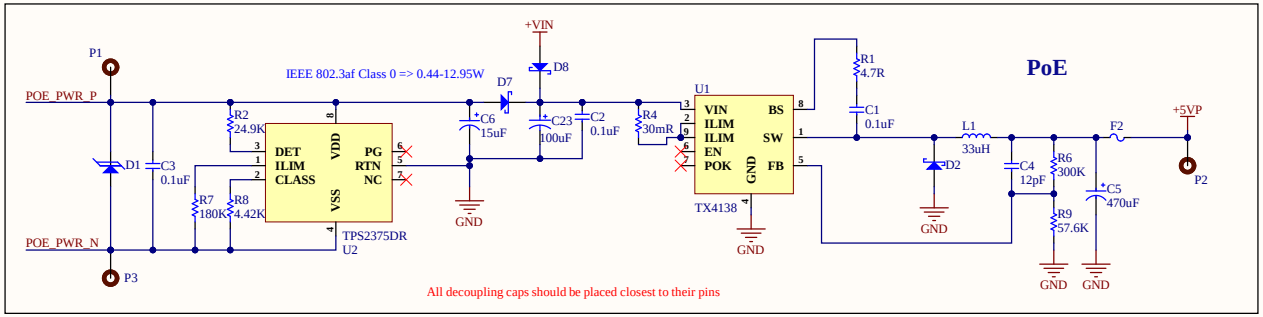
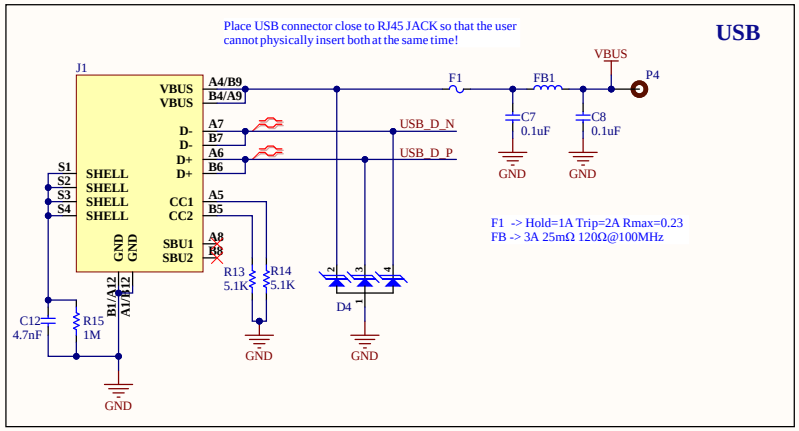
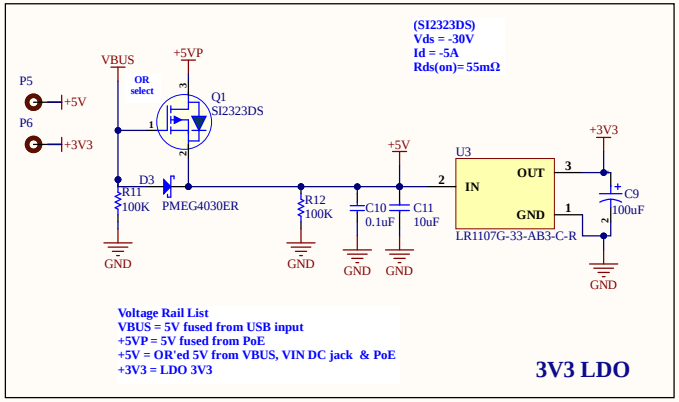
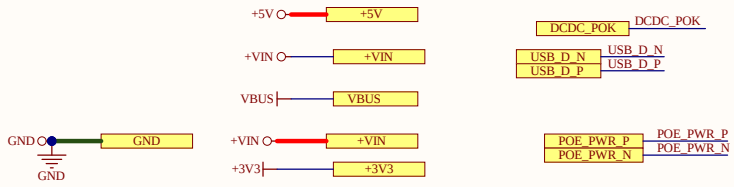


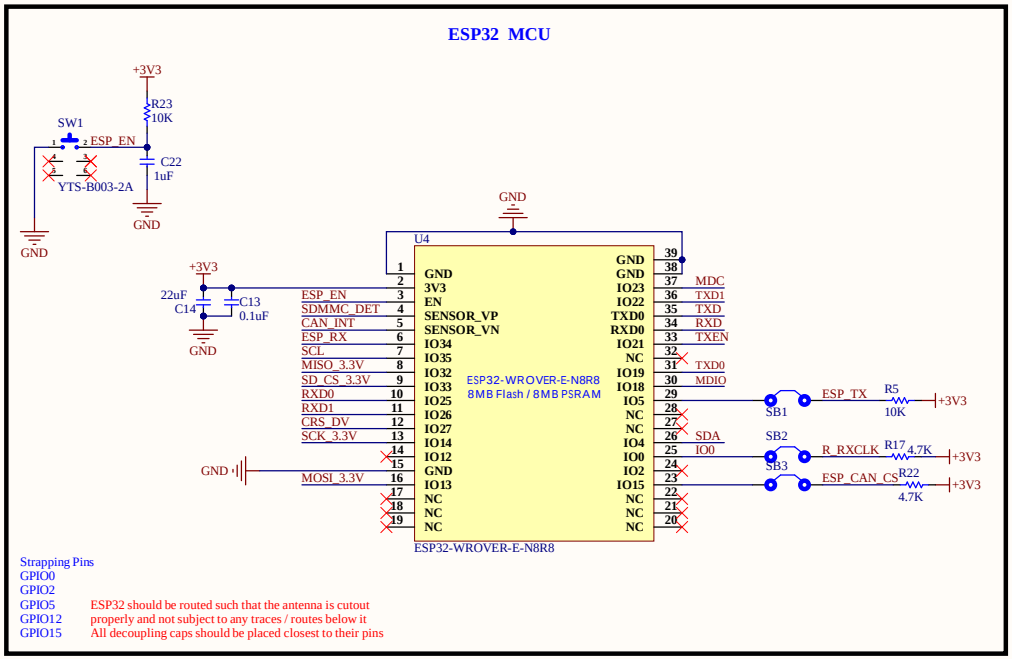






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File: D:\eSpark-Consultants\..\Power Sheet1.SchDoc	Drawn By:	

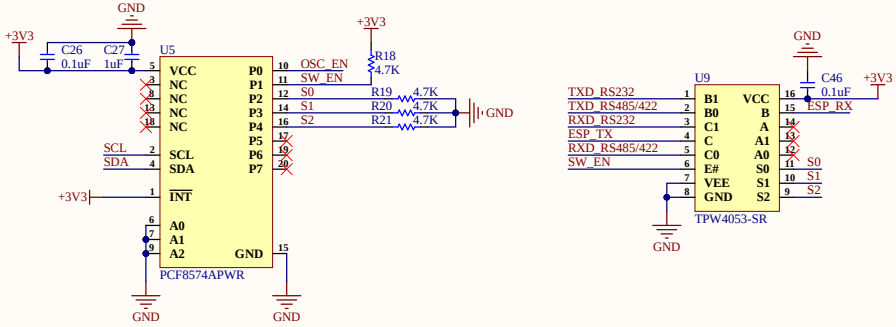
ESP32 MCU



Strapping Pins
GPIO0
GPIO2
GPIO5
GPIO12
GPIO15

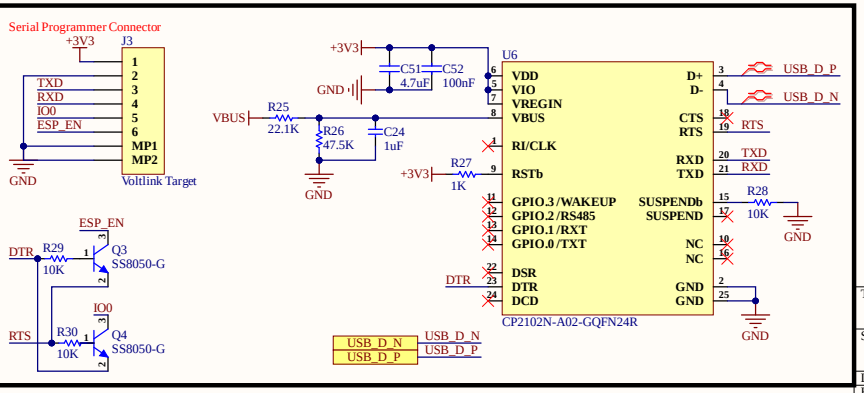
ESP32 should be routed such that the antenna is cutout properly and not subject to any traces / routes below it
All decoupling caps should be placed closest to their pins

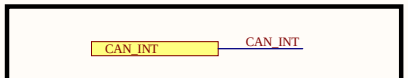
IO expander and Multiplexer

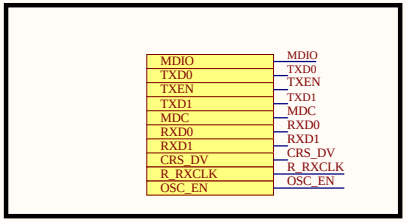


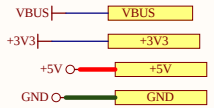
All decoupling caps should be placed closest to their pins

PROGRAMMING



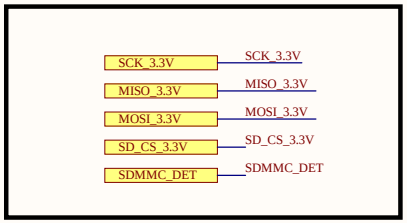


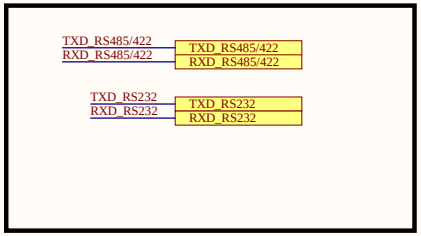




Strapping pins:
* BOOT = IO0
* JTAG = IO3
* VSPI = IO45
* LOG = IO46

ESP32 should be routed such that the antenna is cutout properly and not subject to any traces / routes below it
All decoupling caps should be placed closest to their pins
All debouncing capacitors should be close to their buttons

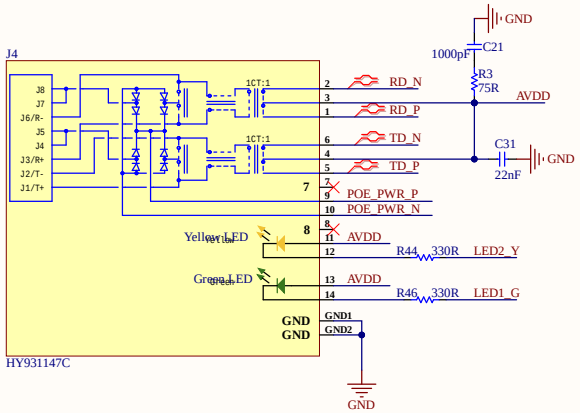


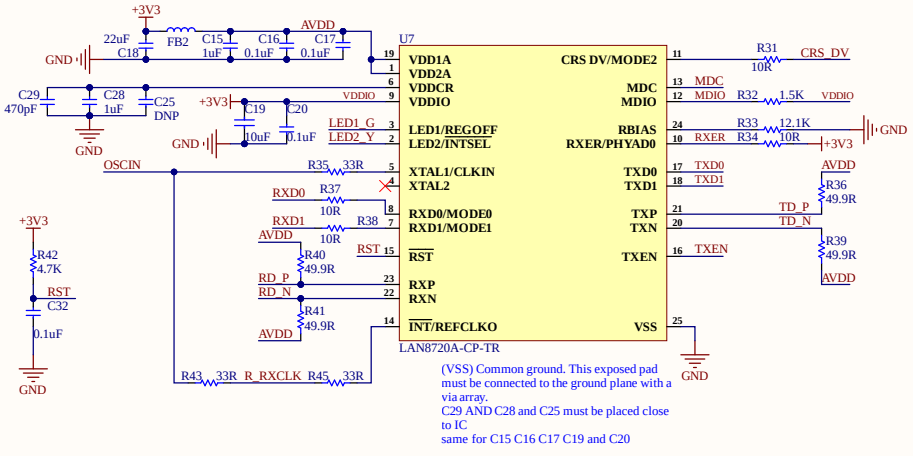


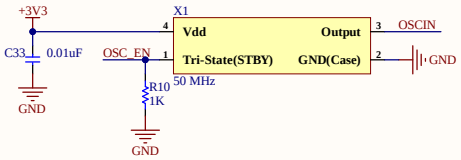
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Size: A3	Number	Revision
Date: 9/23/2024	Sheet of	
File: D:\eSpark-Consultants\..\Controller Sheet.SchDoc	Drawn By:	

RMII 10/100 Ethernet

Route RJ45 RX/TX pairs with 100R diferential trace impedance!







POE_PWR_P
POE_PWR_N

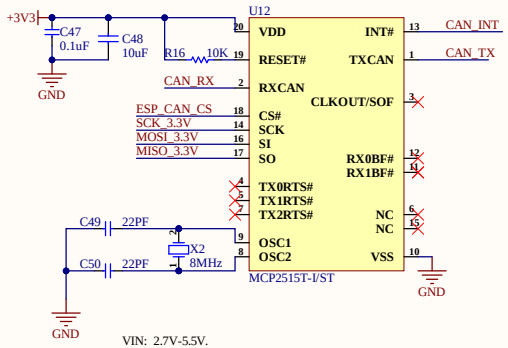
RXD0
RXD1
TXD0
TXD1
TXEN
CRS_DV
MDIO
MDC

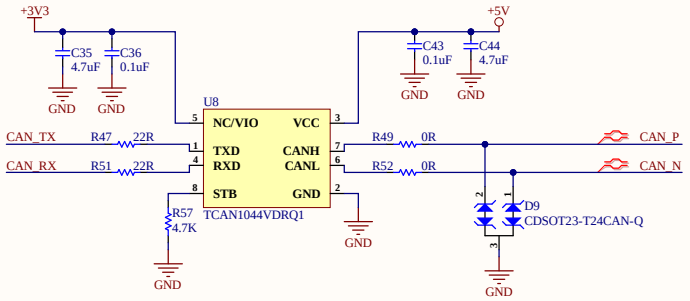
R_RXCLK
OSC_EN

GND
+3V3

Title		
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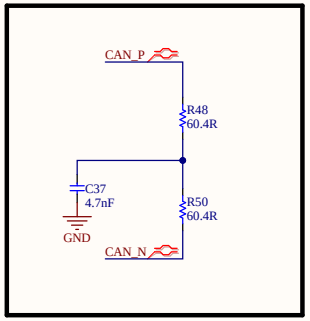
CAN Controller & Transceiver



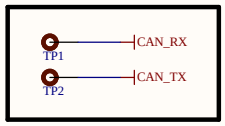


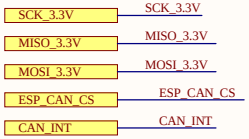
VIN: 2.7V-5.5V.
SPI MODES: 0,0 & 1,1
Temperature Range: -40*C to +85*C

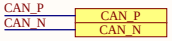
CAN TERMINATION

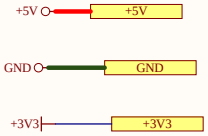


TESTPOINTS



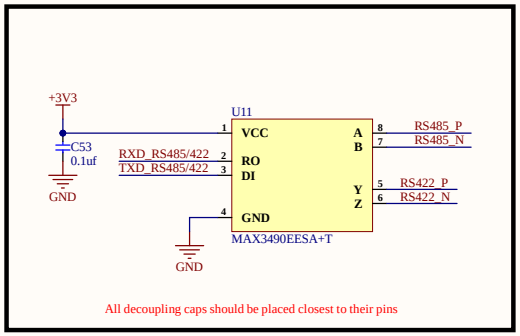




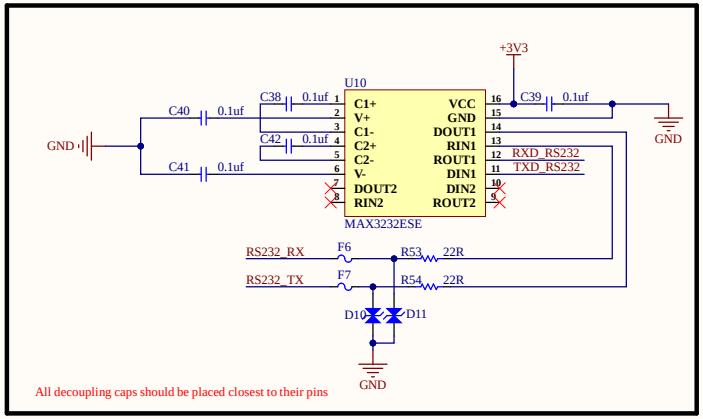


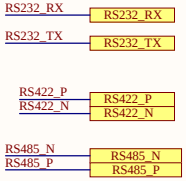
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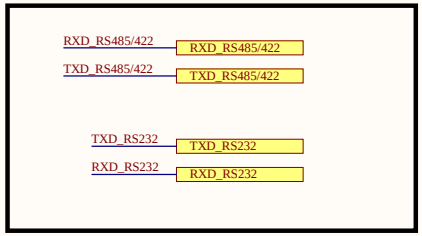
RS485 & RS422

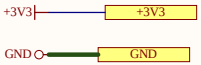


RS232 TO UART



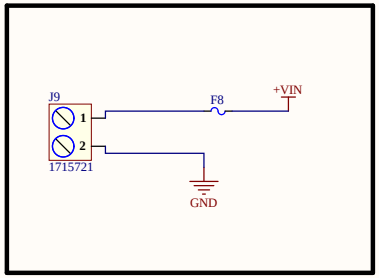




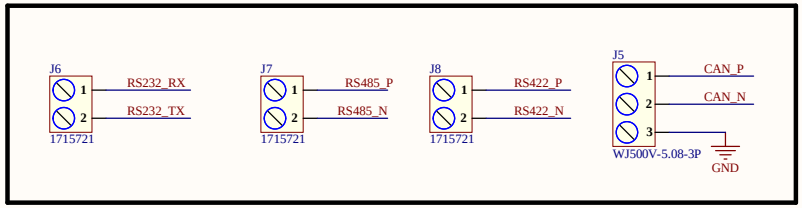


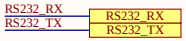
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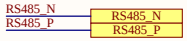
5~60V INPUT



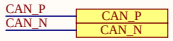
RS-232 RS-485 RS-422 CAN

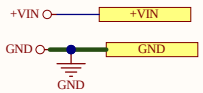






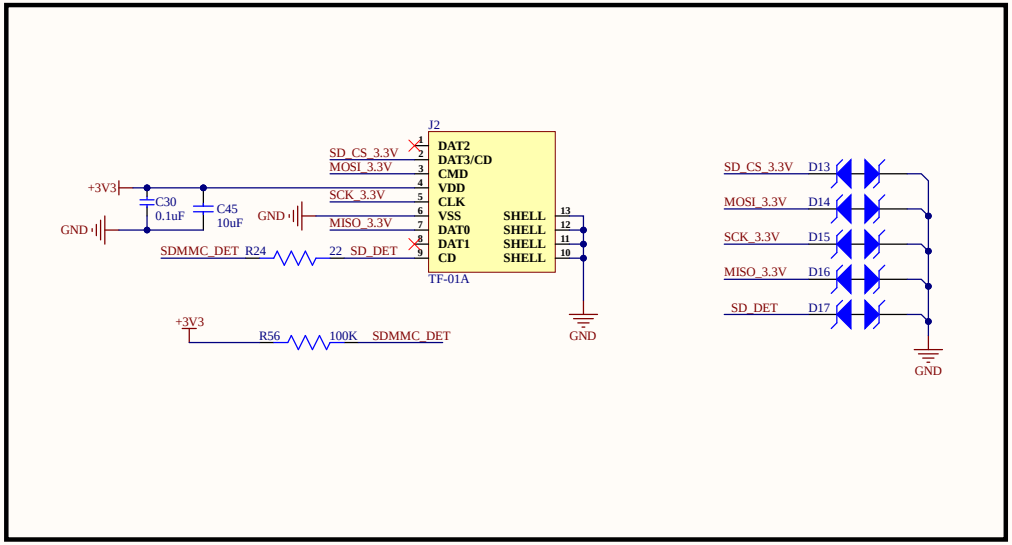






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SD Card Interface



SCK_3.3V
MISO_3.3V
MOSI_3.3V
SD_CS_3.3V
SDMMC_DET

GND
+3V3

Title		
Size: A3	Number	Revision
Date: 9/23/2024	Sheet of	
File: D:\eSpark-Consultants\..\SD Card-Sheet.SchDoc	Drawn By:	